

2025 Clinical Practice Guideline Update by the Infectious Diseases Society of America on the Treatment and Management of COVID-19: Antiviral Treatment for Mild to Moderate COVID-19 in Adults

Amy Hirsch Shumaker,^{1,2,a} Adarsh Bhimraj,^{3,a} Roger Bedimo,⁴ Vincent Chi-Chung Cheng,⁵ Kara W. Chew,⁶ Eric S. Daar,⁷ David V. Glidden,⁸ Robert W. Shafer,⁹ Phyllis C. Tien,¹⁰ Lindsey R. Baden,^{11,12} Kelly Cawcutt,¹³ Kathleen Chiotos,¹⁴ Amy L. Dzierba,¹⁵ Erica J. Hardy,¹⁶ Steven Johnson,¹⁷ Arthur Y. Kim,^{11,18} Christine MacBrayne,¹⁹ Greg S. Martin,²⁰ Nandita Nadig,²¹ Mari M. Nakamura,^{22,23} Carlos R. Oliveira,²⁴ Jeffrey C. Pearson,²⁵ Laura Riley,^{26,27} Shmuel Shoham,²⁸ Pablo Tebas,²⁹ Zachary Willis,³⁰ Joshua Wolf,³¹ Jennifer Loveless,³² Rebecca L. Morgan,^{1,33} Yngve Falck-Ytter,^{1,2,b} and Rajesh T. Gandhi^{11,18,b}

¹Department of Medicine, Case Western Reserve University, School of Medicine, Cleveland, Ohio, USA; ²Department of Medicine, VA Northeast Ohio Healthcare System, Cleveland, Ohio, USA; ³Division of Infectious Diseases, Houston Methodist Hospital, Houston, Texas, USA; ⁴Department of Medicine, UT Southwestern/VA North Texas Health Care System, Dallas, Texas, USA; ⁵Department of Microbiology, Queen Mary Hospital, Li Ka Shing Faculty of Medicine, The University of Hong Kong, Pokfulam, Hong Kong Special Administrative Region, China; ⁶Department of Medicine, David Geffen School of Medicine, University of California Los Angeles, Los Angeles, California, USA; ⁷Department of Medicine, Harbor-UCLA Medical Center, Torrance, California, USA; ⁸Department of Epidemiology and Biostatistics, UCSF, San Francisco, California, USA; ⁹Department of Medicine, Stanford University, Palo Alto, California, USA; ¹⁰Department of Medicine, UCSF/VA San Francisco Health Care System, San Francisco, California, USA; ¹¹Harvard Medical School, Boston, Massachusetts, USA; ¹²Department of Medicine, Brigham and Women's Hospital, Boston, Massachusetts, USA; ¹³Division of Infectious Diseases, University of Nebraska Medical Center, Omaha, Nebraska, USA; ¹⁴Department of Anesthesiology and Critical Care, Children's Hospital of Philadelphia/University of Pennsylvania, Philadelphia, Pennsylvania, USA; ¹⁵Department of Medicine, New York University Grossman School of Medicine, New York City, New York, USA; ¹⁶Departments of Medicine and Obstetrics and Gynecology, Brown University, Providence, Rhode Island, USA; ¹⁷Department of Medicine, University of Colorado School of Medicine, Aurora, Colorado, USA; ¹⁸Department of Medicine, Massachusetts General Hospital, Boston, Massachusetts, USA; ¹⁹Department of Pharmacy, Children's Hospital Colorado, Aurora, Colorado, USA; ²⁰Division of Pulmonary, Allergy, Critical Care and Sleep Medicine, Emory University, Atlanta, Georgia, USA; ²¹Division of Pulmonary and Critical Care Medicine, Northwestern University Feinberg School of Medicine, Chicago, Illinois, USA; ²²Antimicrobial Stewardship Program and Division of Infectious Diseases, Boston Children's Hospital, Boston, Massachusetts, USA; ²³Department of Pediatrics, Harvard Medical School, Boston, Massachusetts, USA; ²⁴Department of Pediatrics, Yale University School of Medicine, New Haven, Connecticut, USA; ²⁵Department of Pharmacy, Brigham and Women's Hospital, Boston, Massachusetts, USA; ²⁶New York-Presbyterian Hospital/Weill Cornell Medical Center, New York City, New York, USA; ²⁷Department of Obstetrics and Gynecology, Weill Cornell Medicine, New York City, New York, USA; ²⁸Division of Infectious Diseases, Johns Hopkins University School of Medicine, Baltimore, Maryland, USA; ²⁹Department of Medicine, University of Pennsylvania, Philadelphia, Pennsylvania, USA; ³⁰Department of Pediatrics, University of North Carolina School of Medicine, Chapel Hill, North Carolina, USA; ³¹Department of Infectious Diseases, St. Jude Children's Research Hospital, Memphis, Tennessee, USA; ³²Clinical Affairs and Practice Guidelines, Infectious Diseases Society of America, Arlington, Virginia, USA; and ³³Department of Health Research Methods, Evidence, and Impact, McMaster University, Hamilton, Ontario, Canada

This article provides a focused update to the clinical practice guideline on the treatment and management of people with COVID-19, developed by the Infectious Diseases Society of America. The guideline panel presents 9 updated recommendations on the use of nirmatrelvir/ritonavir, remdesivir, and molnupiravir, in adults with mild to moderate COVID-19. The recommendations are based on evidence derived from a systematic literature review and adhere to a standardized methodology for rating the certainty of evidence and strength of recommendation according to the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) approach. The panel also provides a section on how to apply these recommendations, including an algorithm on the selection of antivirals.

Keywords. COVID-19; nirmatrelvir/ritonavir; remdesivir; molnupiravir; guideline.

Received 17 November 2025; accepted 21 November 2025; published online 1 April 2026

^aA. H. S. and A. B. contributed equally to this manuscript.

^bY. F.-Y. and R. T. G. contributed equally to this manuscript.

Posted online at <https://www.idsociety.org/practice-guideline/COVID-19-guideline-treatment-and-management/> on 14 October 2025. As COVID-19 treatment and management guidelines may change rapidly with evolving virus variants and ongoing research, please check the website for the most current version of this guideline.

Correspondence: A. Bhimraj, Division of Infectious Diseases, Houston Methodist Hospital, 6560 Fannin St, Ste 1512, Houston, TX 77030, USA (practiceguidelines@idsociety.org).

Clinical Infectious Diseases®

© The Author(s) 2026. Published by Oxford University Press on behalf of Infectious Diseases Society of America. All rights reserved. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com. <https://doi.org/10.1093/cid/ciaf680>

In patients with mild to moderate COVID-19, does treatment with nirmatrelvir/ritonavir versus no treatment result in better outcomes (eg, time to symptom resolution, hospitalization, progression to severe or critical illness)?

Recommendation: In adults with mild to moderate COVID-19 at high risk (Table 1) or with several factors associated with increased risk for progression to severe disease (Table 2), the IDSA guideline panel recommends nirmatrelvir/ritonavir over no antiviral treatment (*strong recommendation, moderate certainty of evidence*).

Remarks:

- Patients' medications should be screened for serious drug interactions with nirmatrelvir/ritonavir.

- Dosing based on renal function:
 - Estimated glomerular filtration rate (eGFR) >60 mL/min: 300 mg nirmatrelvir/100 mg ritonavir every 12 hours for 5 days
 - eGFR <60 mL/min and ≥30 mL/min: 150 mg nirmatrelvir/100 mg ritonavir every 12 hours for 5 days
 - eGFR <30 mL/min including hemodialysis: 300 mg nirmatrelvir/100 mg ritonavir once on day 1 and then 150 mg nirmatrelvir/100 mg ritonavir once daily on days 2–5

Recommendation: In adults with mild to moderate COVID-19 at increased but not high risk for progression to severe disease (Table 2), the IDSA guideline panel suggests using nirmatrelvir/ritonavir over no antiviral treatment (*conditional recommendation, moderate certainty of evidence*).

Remarks:

- Patients who place a higher value on avoiding adverse events and/or drug–drug interactions and a lower value on the small reduction in hospitalization risk or faster symptom resolution may reasonably decline nirmatrelvir/ritonavir.

Recommendation: In adults without risk factors for progression to severe disease (Tables 1 and 2), the IDSA guideline panel suggests against the routine use of nirmatrelvir/ritonavir (*conditional recommendation, moderate certainty of evidence*).

In patients with mild to moderate COVID-19, does treatment with remdesivir versus no treatment result in better outcomes (eg, time to symptom resolution, hospitalization, progression to severe or critical illness)?

Recommendation: In adults with mild to moderate COVID-19 at high risk (Table 1) or with several factors associated with increased risk for progression to severe disease (Table 2), the IDSA guideline panel recommends intravenous remdesivir over no antiviral treatment (*strong recommendation, moderate certainty of evidence*).

Recommendation: In adults with mild to moderate COVID-19 at increased but not high risk for progression to severe disease (Table 2), the IDSA guideline panel suggests intravenous remdesivir over no antiviral treatment (*conditional recommendation, low certainty of evidence*).

Remarks:

- Patients who place a higher value on avoiding daily intravenous infusions for 3 days and a lower value on the small reduction in hospitalization risk may reasonably decline remdesivir.

Recommendation: In adults without risk factors for progression to severe disease (Tables 1 and 2), the IDSA guideline panel suggests against the routine use of remdesivir (*conditional recommendation, moderate certainty of evidence*).

In patients with mild to moderate COVID-19, does treatment with molnupiravir versus no treatment result in better outcomes (eg, time to symptom resolution, hospitalization, progression to severe or critical illness)?

Recommendation: In adults with mild to moderate COVID-19 at high risk (Table 1) or with several factors associated with increased risk for progression to severe disease (Table 2) and who have no other treatment options*, the IDSA guideline panel suggests molnupiravir over no antiviral treatment (*conditional recommendation, low certainty of evidence*).

*Other options for treatment and management of ambulatory patients include 5-day treatment with oral nirmatrelvir/ritonavir or 3-day treatment with intravenous remdesivir.

Remarks:

- See Figure 1 for information from the US Food and Drug Administration (FDA) emergency use authorization (EUA).
- Patients who place a higher value on avoiding reproductive concerns, or the putative mutagenesis, and a lower value on the uncertain benefits would reasonably decline molnupiravir.
- Molnupiravir is not authorized under the FDA EUA for use in patients <18 years because it may affect bone and cartilage growth.
- Molnupiravir is not authorized under the FDA EUA for in-patient use.
- People who engage in sexual activity that may result in conception should use effective contraception during and for 3 months following treatment with molnupiravir. Molnupiravir is not recommended under the FDA EUA for use during pregnancy.

Recommendation: In adults with mild to moderate COVID-19 at increased but not high risk for progression to severe disease (Table 2), the IDSA guideline panel suggests against the routine use of molnupiravir (*conditional recommendation, low certainty of evidence*).

Recommendation: In adults without risk factors for progression to severe disease (Tables 1 and 2), the IDSA guideline panel recommends against the use of molnupiravir (*strong recommendation, moderate certainty of evidence*).

Tables 1 and 2 list examples of risk factors broadly categorized by medical condition or immunosuppressive treatment (partially adapted from the Centers for Disease Control and Prevention [CDC] guidance “Underlying Conditions and the Higher Risk for Severe COVID-19”) [2]. Categorization is based on cohort data from the Omicron era and may not reflect the impact on progression of increasing population immunity or variant characteristics.

The risk of progression to severe COVID-19 is a continuum influenced by various factors, including the severity of risk factor and/or immunosuppression. The categorization of risk and the examples provided in Tables 1 and 2 are illustrative and are not exhaustive or a thorough list of all conditions.

Table 1. Examples of Factors Associated With High Risk of Progression

Example Health Condition	Example Therapeutics ^a
• Age ≥75 y • Fewer than 1% peripheral B-cells assessed in past 6 m • Congenital agammaglobulinemia • Graft-versus-host disease • Hematological malignancy on therapy • HIV infection with CD4 <200 cells/mm³ • Other severe primary immunodeficiency • Solid organ transplant • Solid tumor on immunosuppressive therapy • Stem cell transplant <2 y	• B-cell-depleting agents in the past 12 m (eg, rituximab, ofatumumab, ocrelizumab) • CAR-T therapy in the past 12 m • Abatacept • Tyrosine kinase inhibitor (eg, ibrutinib, acalabrutinib) • High-dose corticosteroids (≥20 mg prednisone or equivalent for ≥4 wks) • Anthracycline derivatives

^aExample therapeutics that predispose to substantial immunosuppression.**Table 2. Examples of Factors Associated With Increased But Not High Risk of Progression^a**

Example Health Condition	Example Therapeutics ^b
• Age 65–74 y • Atherosclerotic cardiovascular disease • Asthma • Cardiomyopathy • Cerebral vascular disease • Chronic liver diseases • Chronic lung diseases • Cirrhosis • Dementia • Diabetes mellitus • Disabilities including Down syndrome • End-stage renal disease • Heart failure • HIV with CD4 >200 cells/mm³ • Inflammatory bowel disease • Mental health conditions • Obesity • Parkinson's disease • Physical inactivity • Pregnancy (current or within 6 wks of pregnancy) • Smoking history • Solid tumor (on immunosuppressive therapy >12 m prior) • Tuberculosis	• Anti-IL-6 • Anti-IL-12 and 23 • Corticosteroids 10–20 mg for ≥4 wks

^aThese factors vary in the degree of risk they confer. For those that confer less risk, the benefit of antiviral therapy will likely be lower.^bExample therapeutics that are less likely to predispose to substantial immunosuppression**BACKGROUND AND APPROACH TO RISK CATEGORIZATION**

Mild to moderate COVID-19 is defined as symptoms of respiratory infection with an oxygen saturation of >94% on room air without the need for supplemental oxygen (Figure 2). Over the past few years, and particularly since December 2021, the risk of severe COVID-19, hospitalization, and death have substantially decreased because of increased immunity and the emergence of the Omicron variant. Therefore, the panel decided to adopt a risk-based approach to recommending antiviral treatment, recognizing that the strength of recommendation will likely be sensitive to varying baseline risks.

As mortality risk among patients hospitalized primarily for COVID-19 has substantially decreased [3], avoiding morbidity leading to hospitalization was the major driver for recommendations on the use of antiviral agents. The CDC reported that cumulative COVID-19-related hospitalization rates dropped in all age groups but remained significantly elevated in those aged 65 years and older [4].

The guideline panel set a threshold for a minimum clinically meaningful reduction of hospitalizations for any antiviral treatment at 10–20 in 1000 (ie, an expected 1–2% fewer hospitalizations from baseline).

To estimate the risk for hospitalization in patients with risk factors for progression to severe disease, the panel relied on event rates from the untreated control arms of recent randomized controlled trials (RCTs) because obtaining homogeneous risk groups is challenging. For example, the PANORAMIC trial [5] started enrollment in December 2021 at the beginning of the Omicron period and appeared to provide a reasonable estimate of hospitalization risk in the placebo group. Despite a substantial proportion of participants in the PANORAMIC trial having risk factors for progression to severe outcomes, only 96 of 12 525 (0.77%) people with COVID-19 required hospitalization. Based on progression rates in PANORAMIC, the panel judged that the group without risk factors for progression to severe disease would have an expected hospitalization rate of 0.5% (or even substantially lower).

Molnupiravir may only be used for the treatment of mild-to-moderate COVID-19 in adults who are at high risk for progression to severe COVID, including hospitalization or death, and for whom alternative COVID-19 treatment options authorized by FDA are not accessible or clinically appropriate.

Figure 1. FDA EUA criteria for the use of molnupiravir [1].

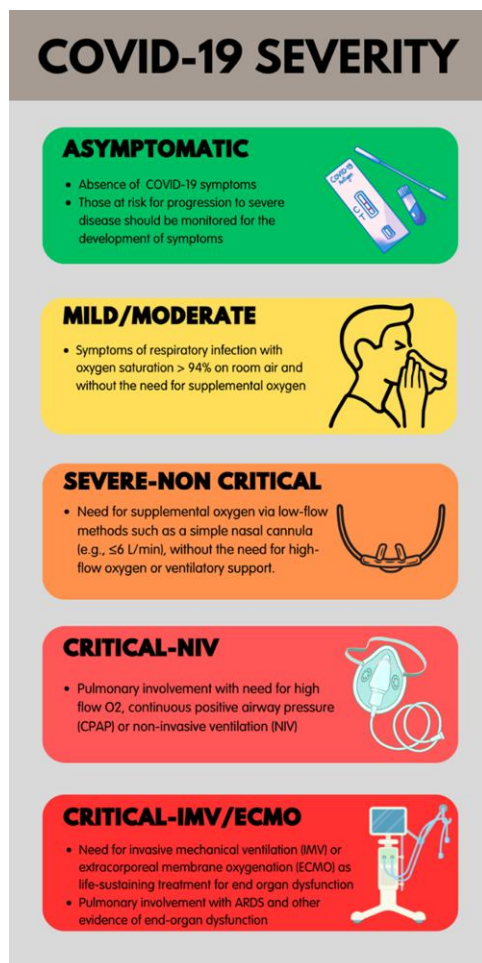


Figure 2. Stages of COVID-19 Severity (Positive antigen or molecular test for SARS-CoV-2). Figure adapted from Shumaker AH, Bhimraj A. Pharmacologic treatment and management of coronavirus disease 2019. *Infect Dis Clin North Am* 2022; 36(2):349–64.

In contrast, numerous medical conditions and immunosuppressive drugs increase the risk for progression to severe disease, of which B-cell-depleting agents and hematologic malignancies are high on the list. It is important to note that the risk of progression to severe COVID-19 is a continuum influenced by various factors, including the degree of immunosuppression. Based on progression rates in cohort studies [6, 7], the panel judged that those at high risk or with several factors associated with increased risk for progression to severe disease would have an expected hospitalization rate of at least 6%. For example, the risk of hospitalization has been reported to be as high as 20–50% among cancer patients [7], though selection bias is difficult to completely exclude.

The panel then judged that those with increased but not high risk for progression to severe disease would have an expected hospitalization rate of >0.5 to <6%; the panel used

the midpoint of this range (3%) for illustrative purposes in the Evidence Profiles. This category is a heterogeneous group, with substantial uncertainty regarding risk for progression. Observational studies are often discordant regarding risk of progression for a particular population (eg, people with chronic liver diseases or those living with mental health conditions) [8]. The group was selected to accommodate conditions and therapeutics that historically have been thought to elevate risk for adverse COVID-19 outcomes but are now considered less likely to cause significant immunosuppression (eg, use of tumor necrosis factor- α inhibitors) [9].

Rationale for Using Anti-SARS-CoV-2 Antivirals

Three direct-acting antivirals are currently available to target SARS-CoV-2 replication at different enzymatic steps:

- **Nirmatrelvir/ritonavir (Paxlovid)** blocks the viral main protease (Mpro), halting polyprotein cleavage and replication. Ritonavir boosts nirmatrelvir levels via CYP3A4 inhibition, allowing twice-daily oral dosing. The agent carries full FDA approval for adults with mild to moderate COVID-19 who are at high risk for progression to severe disease and remains under EUA for adolescents ≥ 12 years (weight ≥ 40 kg) [10, 11].
- **Remdesivir (Veklury)** is a nucleotide analog that binds the viral RNA-dependent RNA polymerase, causing premature chain termination of viral RNA. Initially reserved for in-hospital use, it now has full FDA approval for both hospitalized patients and high-risk outpatients (3-day intravenous [IV] course) across all age groups—from term neonates (≥ 1.5 kg) to adults [12, 13].
- **Molnupiravir (Lagevrio)** is an oral pro-drug converted to β -D-N4-hydroxycytidine (NHC), then phosphorylated to the active triphosphate (NHC-TP) before incorporation into viral RNA by the viral RNA polymerase, inducing lethal mutagenesis. It is available through an EUA for adults ≥ 18 years with mild to moderate COVID-19 who are at high risk for severe outcomes [14].

Most people with mild to moderate COVID-19 cases are managed as outpatients, though some are encountered in emergency or inpatient settings; treatment decisions depend on disease severity, not venue, so the guideline addresses anti-SARS-CoV-2 antivirals by severity rather than setting.

METHODS

The panel's recommendations are based on evidence derived from systematic reviews and adhere to a standardized methodology for rating the certainty of evidence and strength of recommendation according to the Grading of Recommendations,

Assessment, Development, and Evaluation (GRADE) approach (Supplementary Figure 1) [15]. The recommendations have been endorsed by the Society of Critical Care Medicine and the Society for Healthcare Epidemiology of America.

Strong recommendations (“the panel recommends”) are made when the recommended course of action would apply to most people with few exceptions. Conditional recommendations (“the panel suggests”) are made when the suggested course of action would apply to the majority of people with many exceptions and shared decision making is important.

Literature searches were conducted in September 2024 as part of the systematic reviews. Key eligibility criteria at both the topic and clinical question levels guided the selection of studies for inclusion.

A critical appraisal of the evidence according to the GRADE approach, along with an assessment of the benefits and harms of care options, informed the recommendations [15, 16]. Details of the systematic review and guideline development processes are available in the [Supplementary Data](#).

NIRMATRELVIR/RITONAVIR

Summary of Evidence for Nirmatrelvir/Ritonavir

The literature search identified 4 RCTs reporting on treatment of mild to moderate COVID-19 in patients at risk for progression to severe disease with nirmatrelvir/ritonavir [11, 17–20] in ambulatory and hospital settings (Supplementary Table 1). These studies reported on the following outcomes: mortality, hospitalization, serious adverse events (SAEs), adverse events requiring discontinuation of treatment, and symptom resolution (Table 3).

Benefits of Nirmatrelvir/Ritonavir

Though the evidence for prevention of hospital admission by antiviral therapy drove the panel’s recommendations, mortality data from all care settings are also presented to illustrate the downstream effects of antiviral treatment.

Two RCTs showed a relative risk reduction in hospitalizations of 83% (Supplementary Figure 3). For the high-risk patient group, this translates into a robust absolute risk reduction of 50 fewer hospitalizations per 1000 (95% CI: 54 fewer to 41 fewer), exceeding the threshold set by the panel of at least 20 hospitalizations in 1000 (moderate certainty of evidence). In contrast, for patients without risk factors, only 4 fewer hospitalizations in 1000 (95% CI: 4 fewer to 1 fewer) would be expected, limiting the usefulness of antiviral treatment in this population.

The use of nirmatrelvir/ritonavir in high-risk patients may also lead to faster symptom resolution (HR 1.2; 95% CI: 1.07, 1.35) [20].

All-cause mortality through day 28 pooled from 4 trials may be lower in patients receiving nirmatrelvir/ritonavir compared to no nirmatrelvir/ritonavir (Relative risk 0.53; 95% CI: 0.18, 1.58; low certainty of evidence; Supplementary Figure 2). The certainty of evidence was rated down due to few events leading to an imprecise estimate.

Harms of Nirmatrelvir/Ritonavir

Evidence from RCTs in people with mild to moderate COVID-19 receiving nirmatrelvir/ritonavir demonstrate fewer SAEs (RR 0.52; 95% CI: 0.20, 1.35; Supplementary Figure 4), suggesting that most SAEs are related to the underlying illness rather than the antiviral treatment, as well as fewer adverse events (RR 0.75; 95% CI: 0.50, 1.13, Supplementary Figure 5). There was a higher rate of dysgeusia in the nirmatrelvir/ritonavir group compared to placebo, 5.8% versus 0.2%, respectively [18].

Co-administration is contraindicated with agents whose concentrations are markedly altered by nirmatrelvir/ritonavir or with agents that accelerate its metabolism. These interactions can undermine antiviral efficacy and promote resistance, leading to treatment failure. They can also cause SAEs or life-threatening toxicity from elevated levels of concomitant medications (Figures 3 and 4).

Less severe but clinically meaningful drug interactions may also occur when nirmatrelvir/ritonavir is co-administered with other agents. Levels of immunosuppressive agents such as tacrolimus, cyclosporine, or sirolimus can be increased when administered with nirmatrelvir/ritonavir. Although dose adjustments are possible, the narrow therapeutic index, risk of toxicity, and challenges in managing these interactions may make concomitant use inadvisable in many settings. Hormonal contraceptives containing ethinyl estradiol may have reduced effectiveness due to lowered ethinyl estradiol levels when administered with nirmatrelvir/ritonavir. Women of childbearing potential should be counseled to use a back-up, non-hormonal method of contraception for 1 menstrual cycle after stopping nirmatrelvir/ritonavir. Although nirmatrelvir/ritonavir has numerous potential drug-drug interactions, most can be effectively managed through temporary discontinuation, dose adjustment, or substitution of the interacting medication. Therefore, providers should not routinely avoid its use solely due to interaction concerns.

Other Considerations for Nirmatrelvir/Ritonavir

The panel agreed that the overall certainty of the evidence for the treatment of patients was moderate due to concerns about imprecision (Supplementary Table 2). The panel agreed that the benefits are likely to outweigh any potential harms in patients with COVID-19 who are at high risk of severe disease but recognized concerns with drug interactions must be considered.

Table 3. GRADE Evidence Profile: In patients With Mild to Moderate COVID-19, Does Treatment With Nirmatrelvir/Ritonavir Versus No Treatment Result in Better Outcomes (eg, Time to Symptom Resolution, Hospitalization, Progression to Severe or Critical Illness)?

Certainty Assessment				No. of Patients			Effect					
No. of Studies	Study Design	Risk of Bias	Inconsistency	Indirectness	Imprecision	Other Considerations	Nirmatrelvir/ Ritonavir	No Nirmatrelvir/ Ritonavir*	Relative (95% CI)	Absolute (95% CI)	Certainty	Importance
All-cause mortality (follow-up: 28 d)												
4 [11, 17–19]	Randomized trials	Not serious	Not serious	Not serious	Very serious ^a	None	18/1893 (1.0%)	0.1%	RR 0.53 (0.18–1.58)	0 fewer per 1000 (from 1 fewer to 1 more)	⊕⊕○○ Low	CRITICAL
								0.5%		2 fewer per 1000 (from 4 fewer to 3 more)		
								3.0%		14 fewer per 1000 (from 25 fewer to 17 more)		
Hospitalization (follow-up: 28 d)												
2 [11, 18]	Randomized trials	Not serious	Not serious	Not serious ^b	Serious ^a	None	13/1693 (0.8%)	0.5%	RR 0.17 (0.10–0.31)	4 fewer per 1000 (from 5 fewer to 3 fewer)	⊕⊕⊕○○ Moderate	CRITICAL
								3.0%		25 fewer per 1000 (from 27 fewer to 21 fewer)		
								6.0%		50 fewer per 1000 (from 54 fewer to 41 fewer)		
Serious adverse events												
3 [11, 17, 18]	Randomized trials	Not serious	Not serious	Not serious	Serious ^a	None	32/1895 (1.7%)	91/1881 (4.8%)	RR 0.52 (0.20–1.35)	23 fewer per 1000 (from 39 fewer to 17 more)	⊕⊕⊕○○ Moderate	CRITICAL
Adverse events requiring discontinuation of treatment												
3 [11, 17, 18]	Randomized trials	Not serious	Not serious	Not serious	Serious ^a	None	24/1895 (1.3%)	9/1881 (0.5%)	RR 0.75 (0.5–1.13)	26 fewer per 1000 (from 53 fewer to 14 more)	⊕⊕⊕○○ Moderate	Important
Symptom resolution for patients at high risk (follow-up: 28 d) ^c												
1 [20]	Randomized trials	Not serious	Not serious	Not serious	Serious ^d	None	619/970 (63.8%)	566/986 (57.4%)	HR 1.20 (1.07–1.35)	67 more per 1000 (from 25 more to 110 more)	⊕⊕⊕○○ Moderate	Important

Table 3. Continued

Certainty Assessment			No. of Patients				Effect					
No. of Studies	Study Design	Risk of Bias	Inconsistency	Indirectness	Imprecision	Other Considerations	Nirmatrelvir/ Ritonavir	No Nirmatrelvir/ Ritonavir*	Relative (95% CI)	Absolute (95% CI)	Certainty	Importance
Symptom resolution for all patients (follow-up: 28 d) ^e												
2 [18, 20]	Randomized trials	Not serious	Not serious	Not serious	Serious ^f	None	1066/1624 (65.6%)	1036/1620 (64.0%)	RR 1.02 (0.97–1.08)	13 more per 1000 (from 19 fewer to 51 more)	⊕⊕⊕○ Moderate	Important

Abbreviations: CI, confidence interval; HR, hazard ratio; RR, risk ratio.

^aFew events do not meet the optimal information size and suggest fragility in the estimate.^bStudy data from pre-Omicron era; hospitalization rates have significantly declined, potentially affecting the effect estimate resulting in indirectness. Baseline risk indirectness was judged to be less indirect for the lowest and highest risk categories, so the panel decided not to rate down the certainty of evidence.^cMedian time to resolution: NMV/r 16 d (95% CI: 15, 18) versus placebo 19 d (95% CI: 18, 20).^d95% CI crossing threshold of meaningful difference; median time to resolution: 95% CI not completely separated.^eMedian time to resolution: NMV/r 16 d (95% CI: 15, 18) versus placebo 19 d (95% CI: 18, 20) for EPIC-HR; 12 d (95% CI: 11, 13) versus 13 (95% CI: 12, 14) in EPIC-SR.^f95% CI includes no effect as well as small effect.^{*}For mortality and hospitalization, percentages represent illustrative baseline risks for patients without risk factors for progression to severe disease, patients at increased but not high risk, and patients at high risk.

The evidence confirms that using nirmatrelvir/ritonavir early in the disease process confers maximum benefit. It is critical to make a rapid diagnosis and treat ambulatory patients with COVID-19 early in the disease course. Observational studies have shown a similar benefit among vaccinated patients infected with newer variants [22, 23].

Symptomatic Viral Rebound in Patients Treated With and Without Antiviral Agents, Including Nirmatrelvir/Ritonavir

Symptomatic viral rebound has been estimated to occur in 0.8–6.6% of nirmatrelvir/ritonavir-treated patients in various trials, including EPIC-HR (Table 4) [24–27]. Symptom recurrence has also been described with molnupiravir (5.9%) [26] and no antiviral treatment [24, 29]. Symptom recurrence has generally not been associated with hospitalization [30, 31]. No comparative evidence was found on the effect of repeat treatment with nirmatrelvir/ritonavir (on any other direct-acting antivirals) in patients experiencing symptomatic viral rebound after initial antiviral treatment. The effectiveness of repeating the same antiviral in this context remains unknown, particularly with respect to patient-important outcomes such as hospitalization, need for invasive ventilation, or death. Study limitations of observational medical records database studies include misclassifications in admission diagnosis and absence of adequate compliance determination, among others.

Conclusions and Research Needs for Nirmatrelvir/Ritonavir

The guideline panel recommends the use of nirmatrelvir/ritonavir for patients with mild to moderate COVID-19 at high risk for progression to severe disease within 5 days of symptom onset. However, patients with mild to moderate COVID-19 without risk factors (and an estimated hospitalization risk of 0.5% or less) are unlikely to benefit from antiviral treatment. Patients with increased but not high risk may or may not benefit from antiviral treatment, and treatment is only suggested after applying shared decision making weighing the benefits and potential downsides of treatment.

REMDESIVIR

Summary of Evidence for Remdesivir

The literature search identified 5 RCTs (Supplementary Table 1) assessing the outcomes of mortality, hospitalizations for any cause, and COVID-19–related medically attended visits, as well as SAEs (Table 5) [13, 32–35]. One RCT compared treatment with 3 days of IV remdesivir (200 mg on day 1 followed by 100 mg on days 2 and 3) initiated within 7 days of symptom onset to no treatment with remdesivir in unvaccinated patients [13]. The study enrolled patients at high risk for progression (eg, with obesity, diabetes mellitus, hypertension, immune compromise) or age 60 years or older without prior treatment (eg, monoclonal antibodies) who were not expected

- Alpha 1-adrenoreceptor antagonist: alfuzosin
- Antianginal: ranolazine
- Antiarrhythmic: amiodarone, dronedarone, flecainide, propafenone, quinidine
- Anti-gout: colchicine (in patients with renal and/or hepatic impairment)
- Antipsychotics: lurasidone, pimozide
- Benign prostatic hyperplasia agents: silodosin
- Cardiovascular agents: eplerenone, ivabradine
- Ergot derivatives: dihydroergotamine, ergotamine, methylergonovine
- HMG-CoA reductase inhibitors: such as lovastatin, simvastatin (these drugs can be temporarily discontinued to allow nirmatrelvir/ritonavir use)
- Immunosuppressants: voclosporin
- Microsomal triglyceride transfer protein inhibitor: lomitapide
- Migraine medications: eletriptan, ubrogepant
- Mineralocorticoid receptor antagonists: finerenone
- Opioid antagonists: naloxegol
- PDE5 inhibitor: sildenafil (Revatio®) when used for pulmonary arterial hypertension (PAH)
- Sedative/hypnotics: triazolam, oral midazolam
- Serotonin receptor 1A agonist/serotonin receptor 2A antagonist: flibanserin
- Vasopressin receptor antagonists: tolvaptan

Figure 3. Drugs that are primarily metabolized by CYP3A for which elevated concentrations are associated with serious and/or life-threatening reactions* [21]. *Please check drug interactions before initiating nirmatrelvir/ritonavir as not all therapeutic agents or classes with potential interactions are listed; see **Liverpool COVID-19 interactions website**.

to receive oxygen at the time of enrollment (>94% on room air).

Benefits of Remdesivir

Treatment with remdesivir for 3 days in ambulatory patients reduced hospitalizations and COVID-19–related medically attended visits through day 28 (HR 0.28; 95% CI: 0.1, 0.75; low to moderate certainty of evidence, depending on baseline risk; and HR 0.19; 95% CI: 0.07, 0.56; moderate certainty of evidence, respectively). For the high-risk patient group, this translates into 45 fewer hospitalizations per 1000 (95% CI: 54 fewer to 15 fewer), exceeding the threshold set by the panel of at least 20 hospitalizations in 1000 (moderate certainty of evidence). In contrast, for patients without risk factors, only 4 fewer hospitalizations in 1000 (95% CI: 4 fewer to 1 fewer) would be expected, limiting the usefulness of antiviral treatment in this population. COVID-19–related mortality may be lower in patients receiving remdesivir rather than placebo (RR 0.83; 95% CI: 0.57, 1.19; moderate certainty of evidence; [Supplementary Figure 6](#)); however, given the small baseline risk of mortality, the reduction may not be clinically meaningful (absolute effect:

up to 5 fewer per 1000 persons in the highest risk group; 95% CI: from 13 fewer to 6 more). In the PINETREE RCT, no deaths were observed in the treatment or control arms [13].

Harms of Remdesivir

Remdesivir infusions did not appear to be associated with a greater risk of SAEs compared to no remdesivir (RR 0.78; 95% CI: 0.61, 1.00; moderate certainty of evidence; [Supplementary Figure 7](#)).

Other Considerations for Remdesivir

The panel agreed that the overall certainty of evidence for the treatment of people with mild to moderate COVID-19 was low to moderate (depending on baseline risk; [Table 5](#)) due to concerns about imprecision, as less than half of the original projected sample size was enrolled, leading to few events and fragility of the effect estimate [13] ([Supplementary Table 2](#)). However, compared to prior trials, giving remdesivir early in the course of the viral infection appears to have a robust effect within the limitation of a limited sample size. The panel agreed that benefits are likely to outweigh any potential harms in

- Anticancer drugs: apalutamide, enzalutamide
- Anticonvulsant: carbamazepine, phenobarbital, primidone, phenytoin
- Antimycobacterials: rifampin, rifapentine
- Cystic fibrosis transmembrane conductance regulator potentiators: lumacaftor/ivacaftor
- Herbal products: St. John's Wort (*hypericum perforatum*)

Figure 4. Drugs that are strong CYP3A inducers where significantly reduced nirmatrelvir or ritonavir plasma concentrations may be associated with the potential for loss of virologic response and possible resistance. Paxlovid cannot be started immediately after discontinuation of any of the following medications due to the delayed offset of the recently discontinued CYP3A inducer [21].

people with COVID-19 who are at high risk for severe disease. The evidence confirms that using remdesivir early in the disease process confers maximum benefit. It is critical to make a rapid diagnosis and treat ambulatory patients with COVID-19 early in the disease course.

While effective, remdesivir is administered intravenously, which may present logistical challenges in the outpatient setting. These include the need for 3 consecutive days of IV access, potential complications related to IV placement, and higher costs compared to oral antiviral agents. These practical considerations, along with patient preferences and access to infusion facilities, should be part of a shared decision making process between the provider and patient.

No dosage adjustment is needed for remdesivir for any degree of renal dysfunction, including hemodialysis.

Conclusions and Research Needs for Remdesivir

The guideline panel recommends remdesivir for people with mild to moderate disease who are at high risk for severe COVID-19. However, patients with mild to moderate COVID-19 without risk factors (and an estimated hospitalization risk of 0.5% or less) are unlikely to benefit from antiviral treatment. Patients with increased but not high risk may or may not benefit from antiviral treatment, and treatment with remdesivir is only suggested after applying shared decision making weighing the benefits and potential downsides of treatment.

MOLNUPIRAVIR

Summary of Evidence for Molnupiravir

Ten RCTs informed the recommendation for molnupiravir (Supplementary Table 1) [5, 14, 19, 36–42]. The RCTs reported on the effect of treating at least partially vaccinated participants with COVID-19 with either 800 mg of molnupiravir or placebo, evaluating the outcomes of mortality, hospitalization, symptom improvement, and adverse and SAEs (Table 6). In the largest trial (n = 26411), PANORAMIC, 99% of participants had at least 1 COVID-19 vaccine dose with 92–93% having received 3 doses [5]. Two RCTs reported on treatment of

unvaccinated people with COVID-19 with either 800 mg of molnupiravir or placebo for 5 days [38, 39]. In 1 phase III trial (MOVE-OUT trial) reporting on the outcomes of death, hospitalization, and SAEs, people with mild to moderate COVID-19 received either molnupiravir or placebo within 5 days after the onset of symptoms. A phase IIa trial reported on the outcomes of death and SAEs in people with symptom duration <7 days who received molnupiravir or placebo [38].

Benefits of Molnupiravir

Molnupiravir may prevent COVID-19–related hospitalizations, but the pooled relative effect estimate was imprecise (RR 0.86; 95% CI: 0.70, 1.05; moderate to low certainty of evidence depending on assumed baseline risk; Supplementary Figure 8).

For the high-risk patient group, this translates into 8 fewer hospitalizations per 1000 (95% CI: 18 fewer to 3 more; low certainty of evidence).

COVID-19–related mortality may be lower in patients receiving molnupiravir rather than placebo (RR 0.89; 95% CI: 0.68, 1.17; moderate certainty of evidence; Supplementary Figure 9); however, given the small baseline risk of mortality across the available evidence, the reduction in mortality may not be clinically meaningful (absolute effect: up to 3 fewer per 1000 persons in the highest risk group; 95% CI: from 10 fewer to 5 more).

One adequately blinded RCT reported on improvement of COVID-19 symptoms [14]. Fatigue and cough improved more frequently than with placebo (RR for fatigue: 0.73; 95% CI: 0.54, 0.99; moderate certainty of evidence; and RR for cough: 0.86; 95% CI: 0.65, 1.14; moderate certainty of evidence).

Harms of Molnupiravir

Patients treated with molnupiravir may not experience greater SAEs or adverse events than those receiving placebo (RR for SAEs: 0.78; 95% CI: 0.49, 1.24; high certainty of evidence; Supplementary Figure 10 and RR for adverse events: 0.95; 95% CI: 0.78, 1.17; high certainty of evidence; Supplementary Figure 11).

Based on findings from animal reproduction studies, molnupiravir may cause fetal harm when administered during pregnancy [1]. Molnupiravir raises particular concerns in immunocompromised individuals, who may have prolonged viral shedding and impaired clearance, increasing the theoretical risk of treatment-associated viral mutagenesis. Females of childbearing potential should be counseled to use a reliable method of contraception during treatment and for 4 days after the last dose. Breastfeeding is not recommended during treatment with molnupiravir. Lactating individuals may consider interrupting breastfeeding and may consider pumping and discarding breast milk during treatment and for 4 days after the last dose of molnupiravir [43]. Men of reproductive potential who are sexually active with females of childbearing potential should be counseled to use a reliable method of contraception during treatment and for at least 3 months after the last dose of molnupiravir. Molnupiravir is also not recommended in children <18 years of age for the concern of bone growth.

Molnupiravir does not require renal or hepatic dose adjustment.

Other Considerations for Molnupiravir

The panel agreed that the overall certainty of evidence for treatment of people with mild to moderate COVID-19 was low, in favor of molnupiravir treatment in the higher-risk groups (given concerns with imprecision and a relatively small effect), and moderate for the absence of clinically relevant effect in people without risks for progression to severe disease (Supplementary Table 2).

Molnupiravir use presents additional considerations and potential concerns regarding viral mutagenesis in immunocompromised persons and safety in persons of reproductive age, for which more data are needed to quantify such effects. The panel recognizes that alternative treatment options exist with the possibility of greater benefit and fewer safety concerns. The FDA required manufacturers to conduct additional animal studies on the impact of the drug on spermatogenesis and to establish a pregnancy registry if the drug was inadvertently administered during pregnancy [44]. At the time of publication, there are no published pregnancy cases in the registry.

Conclusions and Research Needs for Molnupiravir

The guideline panel suggests the use of molnupiravir for people with mild to moderate COVID-19 at high risk for progression to severe disease who are within 5 days of symptom onset and have no other treatment options, particularly for patients at highest risk such as treatment with B-cell-depleting agents. However, patients with mild to moderate COVID-19 without risk factors (and an estimated hospitalization risk of 0.5% or less) are unlikely to benefit from antiviral treatment, and the guideline panel recommends against using molnupiravir in this situation. People with increased risk (but not at high

risk) may not benefit from antiviral treatment, and the guideline panel suggests against routine treatment with molnupiravir while applying shared decision making weighing the benefits and potential downsides of treatment.

More data are needed on the potential adverse effects of this medication.

How to Apply Guidelines to Management of Ambulatory Patients With Mild to Moderate COVID-19

Patients presenting with COVID-19 should be assessed for disease severity prior to any clinical decision making regarding therapeutic options (Figure 2). Identifying the patient's disease severity is necessary to ensure that the agents are used in the way they were studied for COVID-19 as well as in concordance with FDA EUA (if applicable).

The guideline panel has developed a point-of-care clinical decision aid (Figure 5) to help prescribers identify the most appropriate treatment options for ambulatory patients with COVID-19. Prescribers can apply these guidelines by first identifying the baseline risk for severe outcomes for COVID-19 for their individual patient, such as the need for hospitalization, noninvasive or invasive ventilation, progression to ICU care, or death. Table 1 lists conditions that are considered high-risk factors of progression to severe COVID-19 outcomes, and Table 2 lists other risk factors that may confer increased risk for severe COVID-19 outcomes. Patients may be considered "high risk" if they have 1 factor associated with high risk of progression or several factors associated with increased but not high risk of progression. If the patient does not have a high-risk condition or lacks a risk factor, the patient can be considered without risk factors for progression to severe disease.

When patients present with symptomatic mild to moderate COVID-19, the choice of antiviral therapy depends on several factors. These include overall effectiveness (ie, absolute risk reduction), the certainty of underlying evidence, and ease of administration (eg, need for IV access or hospitalization), as well as patient-specific factors such as symptom duration, renal function, drug-drug interactions, and patient values and preferences.

Currently, there is not sufficient evidence to support combination antiviral treatment of mild to moderate COVID-19. Because nirmatrelvir/ritonavir has high efficacy and is administered orally, it should be the first option considered for ambulatory patients with mild to moderate COVID-19. While molnupiravir lacks clinically significant drug-drug interactions, using it solely for convenience—without first evaluating whether nirmatrelvir/ritonavir or remdesivir can be used—is not recommended.

Ritonavir, one of the components of nirmatrelvir/ritonavir, is associated with numerous drug interactions. While it has been used in HIV management for nearly 2 decades and is supported by well-established guidance, nirmatrelvir/ritonavir is

Table 4. GRADE Evidence Profile: In patients With Mild to Moderate COVID-19 at High Risk for Progression to Severe Disease, Does Treatment With Nirmatrelvir/Ritonavir Versus No Treatment Affect Symptom Recurrence or Viral Rebound After Completion of Course of Nirmatrelvir/Ritonavir?

Certainty Assessment						
No. of Studies	Study Design	Risk of Bias	Inconsistency	Indirectness	Imprecision	Other Considerations
5 [24–28]	Observational studies ^a	Serious ^b	Not serious	Not serious	Not serious	None
Hospitalizations or all-cause deaths						
		Effect				Importance
		- No direct evidence was found investigating the effect of repeat nirmatrelvir/ritonavir (n/r) treatment in patients experiencing symptomatic viral rebound after initial antiviral treatment. - Rates of viral rebound after n/r treatment have been estimated to be in the range of 2.3% (23/990) in the registration trial EPIC-HR to 3.5% (398/11270; Wang 2022) and seen in Wong 2023 at 6.6% (16/242). - Comparative rates of viral rebound have been seen in untreated persons in the registration trial EPIC-HR (17/980; 1.7%) and in Wong 2023 (170/3787; 4.5%). - Molnupiravir rebound has been reported to occur in 5.9% (139/2374; Wang 2022) and seen in data from Wong 2023 (27/563; 4.8%). - Observational evidence showed hospitalization after n/r is infrequent, ranging from 0.11% (6/5287) to 0.4% (2/483) and 0.44% (50/11270) for n/r; and 0.84% for molnupiravir [25, 28] 2 deaths out of 6 patients occurred in those hospitalized in one study. - The effect of repeating the same drug (or another course) after a				⊕○○○ Very low* CRITICAL

Table 4. Continued

Certainty Assessment									
No. of Studies	Study Design	Risk of Bias	Inconsistency	Indirectness	Imprecision	Other Considerations	Effect	Certainty	Importance
							viral rebound is unknown for patient-important outcomes. • Study limitations of observational medical records database studies include misclassifications in admission diagnosis and absence of adequate compliance determination, among others.		
Serious adverse events – not reported									
									Critical

Abbreviations: CI, confidence interval; RR, risk ratio.

^aRates derived from arms of RCTs are observational in nature (and indirect as it relates to the clinical question) as no comparative effectiveness of repeat treatment in viral rebound was found.^bNo comparative effectiveness available.^cCertainty ratings derived from evidence that has not been peer reviewed or published.

often prescribed in settings less familiar with ritonavir—such as primary care offices, urgent care centers, and emergency departments—where complete medication histories may not be readily available. Clinicians should carefully review the patient's full medication list and use tools such as www.COVID19-druginteractions.org/checker, along with pharmacy input, to assess and manage interaction risks at the point of care.

Some patients may have drug-drug interactions that preclude the use of nirmatrelvir/ritonavir. For example, a person with cardiovascular disease and recent coronary intervention may be taking clopidogrel and a statin. While the statin can be held during treatment, clopidogrel should not be co-administered with nirmatrelvir/ritonavir, as it reduces clopidogrel's effectiveness and may lead to poor outcomes such as restenosis. Another group at elevated risk for severe COVID-19 outcomes is transplant recipients on immunosuppressants such as cyclosporine or tacrolimus. Co-administration of nirmatrelvir/ritonavir with these agents can lead to adverse events such as acute kidney injury or neurotoxicity due to their narrow therapeutic index. While transplant centers may be equipped to adjust calcineurin inhibitor doses, such monitoring may not be feasible in the outpatient settings where nirmatrelvir/ritonavir is often prescribed.

When nirmatrelvir/ritonavir is not a suitable option due to drug-drug interactions, remdesivir may be preferred. However, logistical barriers—such as the need for 3 consecutive days of IV infusion—can limit its feasibility. If neither option is available and the patient is at high risk for progression to severe disease (Table 1), molnupiravir may be considered, with clear communication about its uncertain benefits and potential concerns regarding mutagenicity and possible effects on fertility in individuals of childbearing potential.

RESEARCH NEEDS AND FUTURE DIRECTIONS

With the emergence of newer variants and subvariants along with increasing immunity, it is difficult to ascertain the magnitude of risk decline for severe COVID-19 outcomes, which may also differ substantially across risk groups. Data on risk associated with specific risk conditions are heterogeneous and often contradictory. Longitudinal, observational data on morbidity and mortality trends may help inform but are often fraught with confounding variables that limit the certainty of risk estimates. Researchers can help to mitigate some of these concerns by using standard definitions/criteria for high-risk conditions when describing the risk of severe COVID-19 outcomes in their research.

Newer antiviral agents are needed that overcome some of the limitations of current agents, such as drug-drug interactions, the need for intravenous administration, or fertility/mutagenesis concerns. Ease of use is an important factor to ensure

Table 5. GRADE Evidence Profile: In patients With Mild to Moderate COVID-19, Does Treatment With Remdesivir Versus No Treatment Result in Better Outcomes (eg, Time to Symptom Resolution, Hospitalization, Progression to Severe or Critical Illness)?

Certainty Assessment				No. of Patients			Effect					
No. of Studies	Study Design	Risk of Bias	Inconsistency	Indirectness	Imprecision	Other Considerations	Remdesivir	No Remdesivir*	Relative (95% CI)	Absolute (95% CI)	Certainty	Importance
Mortality (follow-up: 28 d)												
5 [13, 32–35]	Randomized trials	Not serious	Not serious	Not serious	Serious ^a	None	56/1545 (3.6%)	0.1%	RR 0.83 (0.57–1.19)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕⊕⊕○ Moderate	CRITICAL
								0.5%		1 fewer per 1000 (from 2 fewer to 1 more)		
							3.0%			5 fewer per 1000 (from 13 fewer to 6 more)		
Hospitalization (all-cause) (follow-up: 28 d)												
1 [13]	Randomized trials	Not serious	Not serious	Not serious ^b	Serious ^c	None	5/279 (1.8%)	0.5%	HR 0.28 (0.10–0.75)	4 fewer per 1000 (from 4 fewer to 1 fewer)	⊕⊕⊕○ Moderate	Critical
								3.0%		22 fewer per 1000 (from 27 fewer to 7 fewer)	⊕⊕○○ Low	
							6.0%			43 fewer per 1000 (from 54 fewer to 15 fewer)	⊕⊕⊕○ Moderate	
COVID-19-related medically attended visits (follow-up: 28 d)												
1 [13]	Randomized trials	Not serious	Not serious	Not serious	Serious ^c	None	4/246 (1.6%)	21/252 (8.3%)	HR 0.19 (0.07–0.56)	67 fewer per 1000 (from 77 fewer to 36 fewer)	⊕⊕⊕○ Moderate	Important
Symptom alleviation												
1 [13]	Randomized trials	Not serious	Not serious	Not serious	Serious ^c	None	23/66 (34.8%)	15/60 (25.0%)	RR 1.60 (0.74–3.48)	150 more per 1000 (from 65 fewer to 620 more)	⊕⊕⊕○ Moderate	Important
Serious adverse events												
4 [13, 32–34]	Randomized trials	Not serious	Not serious	Not serious	Serious ^c	None	98/688 (14.2%)	77/612 (12.6%)	RR 0.78 (0.61–1.00)	28 fewer per 1000 (from 49 fewer to 0 fewer)	⊕⊕⊕○ Moderate	Critical

Abbreviations: CI, confidence interval; HR, hazard ratio; RR, risk ratio.

^a95% CI includes the potential for appreciable benefit but cannot exclude the potential for harm.^bStudy data from pre-Omicron era; hospitalization rates have significantly declined, potentially affecting the effect estimate resulting in indirectness. Baseline risk indirectness was judged to be less indirect for the lowest and highest risk categories, and therefore, the panel decided not to rate down the certainty of evidence.^cFew events do not meet the optimal information size and suggest fragility in the estimate. For the outcome of hospitalization: at baseline risk of 0.5%, the entire 95% CI of the absolute risk difference is within the no effect boundary and only fragility is present (rated down once); at baseline risk of 3%, the 95% CI of the absolute risk difference includes appreciable benefit as well as little or no effect on hospitalization: rated down twice; at 6% baseline risk, the entire 95% CI of the absolute risk reduction is above the panel determination of minimally important difference (rated down once).^{*}For mortality and hospitalization, percentages represent illustrative baseline risks for patients without risk factors for progression to severe disease, patients at increased but not high risk, and patients at high risk.

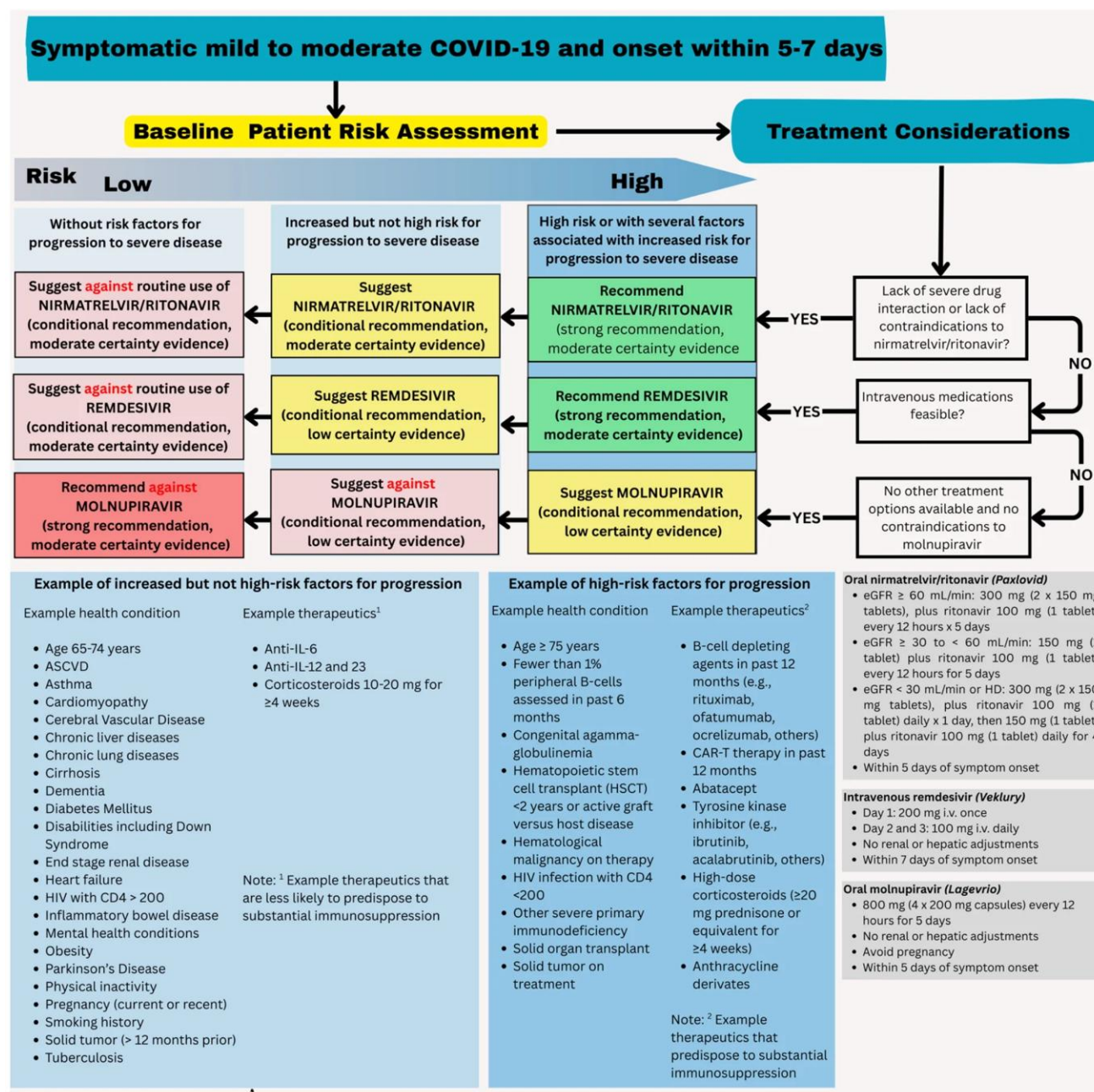


Figure 5. Approaches to treatment of mild to moderate COVID-19.

antiviral agents are appropriately utilized as evidence suggests that current antiviral agents are underprescribed in at-risk populations [45, 46].

Supplementary Data

Supplementary materials are available at [Clinical Infectious Diseases](https://academic.oup.com/cid/advance-article/doi/10.1093/cid/ciaf680/8571315) online. Consisting of data provided by the authors to benefit the reader, the posted materials are not copyrighted and are the sole responsibility of the authors, so questions or comments should be addressed to the corresponding author.

Notes

Acknowledgments. The expert panel acknowledges former panelists for their work on the primary guideline. The panel acknowledges the following organizations and selected reviewers for their review of the draft manuscript: Pediatric Infectious Diseases Society, Society of Infectious Diseases Pharmacists, Society for Healthcare Epidemiology, and Society for Critical Care Medicine; Drs. Jason Gallagher and Roy Gulick; and

Drs. Melissa Badowski and Melissa Nolan on behalf of the Infectious Diseases Society of America (IDSA) Standards & Practice Guidelines Subcommittee.

Drs. Adarsh Bhimraj and Rajesh T. Gandhi are chair and vice chair of the panel, respectively. The Nonhospitalized Patients subgroup, under the leadership of Dr. Amy Shumaker, led the development of the recommendations. Remaining panelists assisted with interpretation of data, as well as drafting, revising, and approving the recommendations and manuscript. Drs. Yngve Falck-Ytter, lead methodologist, and Rebecca Morgan, methodologist, were responsible for designing and performing the data analyses and leading the panel according to the GRADE process. Jennifer Loveless, methodologist, was responsible for project planning and management, including revisions to and final approval of the recommendations and manuscript.

Disclaimer. It is important to recognize that guidelines cannot always account for individual variation among patients. They are assessments of current scientific and clinical information provided as an educational service; are not continually updated and may not reflect the most recent evidence (new evidence may emerge between the time information is drafted and when it is published or read); should not be considered inclusive of all proper methods of care, or as a statement of the standard of care; do not mandate any course of medical care; and are not intended to supplant clinician judgment with respect to particular patients or situations. Whether to follow guidelines and to what extent is voluntary, with the ultimate determination regarding their application to be made by the clinician in light of each patient's individual circumstances. While IDSA makes every effort to present accurate, complete, and reliable information, these guidelines are presented "as is" without any warranty, either express or implied. IDSA (and its officers, directors, members, employees, and agents) assume no responsibility for any loss, damage, or claim with respect to any liabilities, including direct, special, indirect, or consequential damages, incurred in connection with these guidelines or reliance on the information presented.

The guidelines represent the proprietary and copyrighted property of IDSA. All rights reserved. No part of these guidelines may be reproduced, distributed, or transmitted in any form or by any means, including photocopying, recording, or other electronic or mechanical methods, without the prior written permission of IDSA. Permission is granted to physicians and health care providers solely to copy and use the guidelines in their professional practices and clinical decision-making. No license or permission is granted to any person or entity, and prior written authorization by IDSA is required to sell, distribute, or modify the guidelines, or to make derivative works of or incorporate the guidelines into any product, including, but not limited to, clinical decision support software or any other software product. Except for the permission granted above, any person or entity desiring to use the guidelines in any way

must contact IDSA for approval in accordance with the terms and conditions of third-party use, in particular any use of the guidelines in any software product.

Financial support. This work was supported by the Infectious Diseases Society of America. This project has been funded in whole or in part with federal funds from the National Cancer Institute, National Institutes of Health, under contract no. 75N91019D00024. The content of this publication does not necessarily reflect the views or policies of the Department of Health and Human Services, nor does mention of trade names, commercial products, or organizations imply endorsement by the US government.

Potential conflicts of interest. Evaluation of relationships as potential conflicts of interest (COIs) is determined by a review process. The assessment of disclosed relationships for possible COIs is based on the relative weight of the financial relationship (ie, monetary amount) and the relevance of the relationship (ie, the degree to which an association might reasonably be interpreted by an independent observer as related to the topic or recommendation of consideration). The following panelists have scientific advisory/consultant roles *not related to the topic of COVID-19* with indicated companies: R. B. with Merck and Gilead, E. S. D. with Gilead, D. V. G. with Merck and Gilead, S. S. with Pfizer (concluded), and P.T. with Merck. The following panelists have scientific advisory/consultant roles related to outpatient antivirals but not nirmatrelvir/ritonavir, remdesivir, or molnupiravir with indicated companies: R. B. with Shionogi, K. W. C. with Pardes Biosciences (concluded), A. Y. K. with Shionogi, S. S. with Adamis and Immunome (concluded), and P. T. with Shionogi. K. W. C. and E. S. D. have non-financial COIs related to nirmatrelvir/ritonavir and ensitrelvir, respectively. K. W. C.'s participation was carefully managed during discussions, and she recused from all voting. No disclosures were reported for all other authors (the majority of panelists), including chair and vice chair.

All authors have submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Conflicts that the editors consider relevant to the content of the manuscript have been disclosed.

Additional information. More detailed information on the analysis and development of the recommendation is available in the [Supplementary Material](#).

References

1. U.S. Food and Drug Administration. Fact sheet for patients and caregivers: Emergency Use Authorization (EUA) of Molnupiravir for Coronavirus Disease 2019 (COVID-19). Available at: <https://www.fda.gov/media/155055/download>. Accessed 13 February 2023.
2. Centers for Disease Control and Prevention. People with certain medical conditions and COVID-19 risk factors. Available at: <https://www.cdc.gov/covid/risk-factors/index.html>. Accessed 12 March 2025.
3. Adjei S, Hong K, Molinari N-AM, et al. Mortality risk among patients hospitalized primarily for COVID-19 during the Omicron and Delta variant pandemic periods—United States, April 2020–June 2022. *MMWR Morb Mortal Wkly Rep* 2022; 71: 1182–9.

4. Taylor CA, Patel K, Pham H, et al. COVID-19-associated hospitalizations among U.S. adults aged ≥ 18 years—COVID-NET, 12 states, October 2023–April 2024. *MMWR Morb Mortal Wkly Rep* **2024**; 73:869–75.
5. Butler CC, Hobbs FDR, Gbinigie OA, et al. Molnupiravir plus usual care versus usual care alone as early treatment for adults with COVID-19 at increased risk of adverse outcomes (PANORAMIC): an open-label, platform-adaptive randomised controlled trial. *Lancet* **2023**; 401:281–93.
6. Evans RA, Dube S, Lu Y, et al. Impact of COVID-19 on immunocompromised populations during the Omicron era: insights from the observational population-based INFORM study. *Lancet Reg Health Eur* **2023**; 35:100747.
7. Pinato DJ, Patel M, Scotti L, et al. Time-dependent COVID-19 mortality in patients with cancer: an updated analysis of the OnCovid Registry. *JAMA Oncol* **2022**; 8:114–22.
8. Zsichla L, Müller V. Risk factors of severe COVID-19: a review of host, viral and environmental factors. *Viruses* **2023**; 15:175.
9. Izadi Z, Brenner EJ, Mahil SK, et al. Association between tumor necrosis factor inhibitors and the risk of hospitalization or death among patients with immune-mediated inflammatory disease and COVID-19. *JAMA Netw Open* **2021**; 4:e2129639.
10. U.S. Food and Drug Administration. Fact sheet for healthcare providers: emergency use authorization for Paxlovid™. Available at: <https://www.fda.gov/media/155050/download>. Accessed 22 December 2021.
11. Hammond J, Leister-Tebbe H, Gardner A, et al. Oral nirmatrelvir for high-risk, nonhospitalized adults with COVID-19. *N Engl J Med* **2022**; 386:1397–408.
12. Gordon CJ, Tchesnokov EP, Woolner E, et al. Remdesivir is a direct-acting antiviral that inhibits RNA-dependent RNA polymerase from severe acute respiratory syndrome coronavirus 2 with high potency. *J Biol Chem* **2020**; 295:6785–97.
13. Gottlieb RL, Vaca CE, Paredes R, et al. Early remdesivir to prevent progression to severe COVID-19 in outpatients. *N Engl J Med* **2022**; 386:305–15.
14. Guan Y, Puenpatom A, Johnson MG, et al. Impact of molnupiravir treatment on patient-reported COVID-19 symptoms in the Phase 3 MOVE-OUT trial: a randomized, placebo-controlled trial. *Clin Infect Dis* **2023**; 77:1521–30.
15. Guyatt GH, Oxman AD, Vist GE, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendation. *BMJ* **2008**; 336:924–6.
16. Infectious Diseases Society of America. IDSA handbook on clinical practice guideline development. Available at: <https://www.idsociety.org/practice-guideline/clinical-practice-guidelines-development-training-and-resources/>. Accessed 21 April 2025.
17. Liu J, Pan X, Zhang S, et al. Efficacy and safety of Paxlovid in severe adult patients with SARS-CoV-2 infection: a multicenter randomized controlled study. *Lancet Reg Health West Pac* **2023**; 33:100694.
18. Hammond J, Fountaine RJ, Yunis C, et al. Nirmatrelvir for vaccinated or unvaccinated adult outpatients with COVID-19. *N Engl J Med* **2024**; 390:1186–95.
19. RECOVERY Collaborative Group; Horby PW, Staplin N, et al. Molnupiravir or nirmatrelvir-ritonavir versus usual care in patients admitted to hospital with COVID-19 (RECOVERY): a randomised, controlled, open-label, platform trial. *Lancet Infect Dis* **2025**; 25:1000–10.
20. Hammond J, Leister-Tebbe H, Gardner A, et al. Alleviation of COVID-19 Symptoms and reduction in healthcare utilization among high-risk patients treated with nirmatrelvir/ritonavir (NMV/R): a phase 3 randomized trial. *Clin Infect Dis* **2025**; 80:323–30.
21. Paxlovid. Prescribing information. Pfizer, US. Available at: <https://labeling.pfizer.com/ShowLabeling.aspx?id=19599>. Accessed 28 March 2025.
22. Arbel R, Sagy YW, Hoshen M, et al. Nirmatrelvir use and severe COVID-19 outcomes during the Omicron surge. *N Engl J Med* **2022**; 387:790–8.
23. Wong CKH, Au ICH, Lau KTK, et al. Real-world effectiveness of molnupiravir and nirmatrelvir plus ritonavir against mortality, hospitalisation, and in-hospital outcomes among community-dwelling, ambulatory patients with confirmed SARS-CoV-2 infection during the Omicron wave in Hong Kong: an observational study. *Lancet* **2022**; 400:1213–22.
24. Anderson AS, Caubel P, Rusnak JM; EPIC-HR Trial Investigators. Nirmatrelvir-ritonavir and viral load rebound in COVID-19. *N Engl J Med* **2022**; 387:1047–9.
25. Ranganath N, O'Horo JC, Challener DW, et al. Rebound phenomenon after nirmatrelvir/ritonavir treatment of Coronavirus disease 2019 (COVID-19) in high-risk persons. *Clin Infect Dis* **2023**; 76:e537–9.
26. Wang L, Berger NA, Davis PB, Kaelber DC, Volkow ND, Xu R. COVID-19 rebound after Paxlovid and Molnupiravir during January–June 2022. *medRxiv* 22276724 [Preprint]. 22 June 2022. Available from: <https://doi.org/10.1101/2022.06.21.22276724>.
27. Wong CKH, Lau KTK, Au ICH, et al. Viral burden rebound in hospitalised patients with COVID-19 receiving oral antivirals in Hong Kong: a population-wide retrospective cohort study. *Lancet Infect Dis* **2023**; 23:683–95.
28. Malden DE, Hong V, Lewin BJ, et al. Hospitalization and emergency department encounters for COVID-19 after Paxlovid treatment—California, December 2021–May 2022. *MMWR Morb Mortal Wkly Rep* **2022**; 71:830–3.
29. U.S. Food and Drug Administration. FDA briefing document. Available at: <https://www.fda.gov/media/166197/download>. Accessed 20 March 2023.
30. Harrington PR, Cong J, Troy SB, et al. Evaluation of SARS-CoV-2 RNA rebound after nirmatrelvir/ritonavir treatment in randomized, double-blind, placebo-controlled trials—United States and international sites, 2021–2022. *MMWR Morb Mortal Wkly Rep* **2023**; 72:1365–70.
31. Smith DM, Li JZ, Moser C. Recurrence of symptoms following a 2-day symptom free period in patients with COVID-19. *JAMA Netw Open* **2022**; 5:e2238867.
32. Beigel JH, Tomashek KM, Dodd LE, et al. Remdesivir for the treatment of COVID-19—final report. *N Engl J Med* **2020**; 383:1813–26.
33. Sise ME, Santos JR, Goldman JD, et al. Efficacy and safety of remdesivir in people with impaired kidney function hospitalized for COVID-19 pneumonia: a randomized clinical trial. *Clin Infect Dis* **2024**; 79:1172–81.
34. Spinner CD, Gottlieb RL, Criner GJ, et al. Effect of remdesivir vs standard care on clinical status at 11 days in patients with moderate COVID-19: a randomized clinical trial. *JAMA* **2020**; 324:1048–57.
35. WHO Solidarity Trial Consortium. Repurposed antiviral drugs for COVID-19—interim WHO SOLIDARITY trial results. *N Engl J Med* **2021**; 384:497–511.
36. Arribas JR, Bhagani S, Lobo SM, et al. Randomized trial of molnupiravir or placebo in patients hospitalized with COVID-19. *NEJM Evid* **2022**; 1:EVIDoa2100044.
37. Caraco Y, Crofoot GE, Moncada PA, et al. Phase 2/3 trial of molnupiravir for treatment of COVID-19 in nonhospitalized adults. *NEJM Evid* **2022**; 1:EVIDoa2100043.
38. Fischer WA 2nd, Eron JJ Jr, Holman W, et al. A phase 2a clinical trial of molnupiravir in patients with COVID-19 shows accelerated SARS-CoV-2 RNA clearance and elimination of infectious virus. *Sci Transl Med* **2021**; 14:eab17430.
39. Jayk Bernal A, Gomes da Silva MM, Musungaie DB, et al. Molnupiravir for oral treatment of COVID-19 in nonhospitalized patients. *N Engl J Med* **2021**; 386:509–20.
40. Khoo SH, FitzGerald R, Saunders G, et al. Molnupiravir versus placebo in unvaccinated and vaccinated patients with early SARS-CoV-2 infection in the UK (AGILE CST-2): a randomised, placebo-controlled, double-blind, phase 2 trial. *Lancet Infect Dis* **2023**; 23:183–95.
41. Sinha S, Kumarasamy N, Suram VK, et al. Efficacy and safety of molnupiravir in mild COVID-19 patients in India. *Cureus* **2022**; 14:e31508.
42. Zou R, Peng L, Shu D, et al. Antiviral efficacy and safety of molnupiravir against Omicron variant infection: a randomized controlled clinical trial. *Front Pharmacol* **2022**; 13:939573.
43. U.S. Food and Drug Administration. Frequently asked questions on the emergency use authorization for Lagevrio (molnupiravir) for treatment of COVID-19. Available at: <https://www.fda.gov/media/155056/download>. Accessed 13 February 2023.
44. Lagevrio FAQs. Available at: https://www.lagevriohcp.com/frequently-asked-questions/?utm_source=bing&utm_medium=cpc&utm_campaign=Lagevrio%20HCP_Generic_General_BRND_NA_ENGM_EXCT_TEXT_NA&utm_term=molnupiravir_eua&utm_content=EUA&gclid=878f8e846bf61c2f0821e9734e8aa0e3&gclid=878f8e846bf61c2f0821e9734e8aa0e3. Accessed 12 July 2025.
45. Monach PA, Anand ST, Fillmore NR, et al. Underuse of antiviral drugs to prevent progression to severe COVID-19—Veterans Health Administration, March–September 2022. *MMWR Morb Mortal Wkly Rep* **2024**; 73:57–61.
46. Wilcock AD, Kissler S, Mehrotra A, et al. Clinical risk and outpatient therapy utilization for COVID-19 in the Medicare population. *JAMA Health Forum* **2024**; 5:e235044.